Ayyaz Akhtar

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EDUCATION

MS Cybersecurity
New York University
GPA: 3.90 - (2023 – 2025)

MS Information Technology
Virginia Tech. University
GPA: 3.97 - (2014 – 2017)

BS Information Technology
George Mason University
GPA: 3.32 - (2011 – 2014)

Technologies

Client
HTML, JavaScript, TypeScript

Server
Nodejs, Bash, Python, Java, Go

Database
SQL, Doc., Col., KeyVal, Graph

Data Analytics
Data Lakes, Data Warehouse

Artificial Intelligence
Agents, Training, Inferencing

Cloud
AWS, Azure, GCP, On-Prem

Compute
Kubernetes, Serverless, HPC

Security
PKI, SIEM, EDR, SSO, IAM, RBAC

Networking
DNS, VPC, VPN, VLAN, ACL, BGP

CI/CD
GHA, Argro, Kargo, Jenkins

DevOps
Helm, Terraform, Ansible, Vault

Monitoring
Logs, Metrics, Tracing, Alerts

Summary

Hands-on engineer with 11+ years of experience architecting and delivering distributed systems, security infrastructure, and developer platforms at enterprise scale. Deeply interested in security engineering, infrastructure automation, and thoughtful system design that balances performance, resilience, and operational simplicity.

Projects

Secure Multi-Cloud Hybrid Infrastructure
Designed and implemented a secure multi-cloud hybrid infrastructure leveraging VPN tunneling, network segmentation, two-factor authentication, and comprehensive host and service monitoring.

Wireless Network Security Research
Investigated and simulated wireless network attacks including deauthentication floods, MAC spoofing, and rogue AP injection to identify exploitable vulnerabilities and assess the effectiveness of countermeasures.

Predictive Analytics & Graph Theory
Applied graph theory and data mining techniques to detect communities, optimize recommendations, and model potential transmission pathways across networks.

EXPERIENCE

Senior Security Engineer
Coreweave | New York, NY (2025 – Present)

Built and operationalized an on-premises Elasticsearch SIEM capable of ingesting 20TB+ of daily security telemetry, deploying the full stack through GitOps automation (Argo CD, Kargo, GitHub Actions) and Infrastructure as Code (Terraform, Helm).

- Developed a scalable Terraform module framework to standardize and accelerate resource provisioning across the platform regions and environments.
- Established the observability foundation for the SIEM platform, integrating Grafana, Loki, Prometheus, and Alertmanager to deliver centralized visibility and automated incident escalation through Slack and PagerDuty.
- Implemented bi-directional cross-cluster replication (CCR) across Elasticsearch clusters to ensure high availability and automated failover, eliminating single points of failure for the Siem.

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Homelab

Infrastructure
pfsense - firewall
unifi - vlan
technitium - dns
proxmox - hypervisor
unraid - nas

Kubernetes Cluster
Nodes - 3 control/6 worker
Services - 40+
CPU - 84 cores
RAM - 216 GB
Uplink - 2.5 Gb
Disk - 8 TB
NAS - 88 TB

Certifications

Amazon Web Services
DevOps Engineer
Solutions Architect

Scrum Alliance
Scrum Master
Scrum Developer
Scrum Product Owner

CompTIA
Network+
Security+
PenTest+
CySA+
CASP+

Github

aakhtar3
aakhtar3.github.io
javascript-projects
school-projects

disneystreaming
automated-cloud-advisor

osirislab
CSAW-CTF-2023-Finals

EXPERIENCE – CONTINUED

Lead Software Engineer
The Walt Disney Company | New York, NY (2019 – 2025)

Served as technology lead across product, platform, and engineering organizations, driving architectural strategy and delivering cohesive web infrastructure for millions of customers.

- Increased web throughput by 70% by optimizing caching layers, resolving bandwidth limitations, and eliminating performance bottlenecks across critical infrastructure.
- Configured IP-based access control using edge cloudlets and implemented custom redirect rules to enforce secure access and dynamic traffic handling across environments.
- Streamlined subdomain structure to support secure, PCI-compliant integration with external systems, reducing complexity and improving cross-origin communication

Senior Software Engineer
Capital One | McLean, VA (2016 – 2019)

Contributed to enterprise-scale web and cloud infrastructure at Capital One, delivering full-stack solutions across frontend, CI/CD, and serverless architecture while driving developer experience improvements and cost optimization.

- Made significant contributions to Capital One's enterprise Node.js framework, driving consistency, reusability, and improved developer experience across internal applications.
- Upgraded cloud application architecture to support secure, scalable, and compliant CI/CD pipelines aligned with enterprise delivery standards.

Software Engineer
Booz Allen Hamilton | McLean, VA (2014 – 2016)

Operated as a full-stack engineer in a test-driven environment, delivering scalable intelligence platforms and security tooling that supported real-time data analysis, threat reporting, and mission-critical decision-making.

- Revamped a legacy security platform's API by remediating critical vulnerabilities, implementing modern authentication, and enabling end-to-end encryption.
- Developed internal tooling enabling analysts to access, filter, and visualize near real-time threat intelligence reports, improving response times and decision-making.